Harpuneet Singh

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Education

University of Toronto, *HBSc Computer Science (Co-op)* Aug 2023 – Jul 2027
Toronto, Ontario, Canada
Recognitions: Dr. Babak Farzad Award for Leadership, Laidlaw Research Scholar, PRISM Fellow, Dean's List Scholar

Learning Paths High School, *High School Diploma* Jul 2021 – May 2023
Ajitgarh, Punjab, India

St Kabir Public School, *Secondary School Diploma* Jul 2011 – Jul 2021
Chandigarh, India

Technical Skills

Languages: C++, Python, C, Verilog, MicroPython, Assembly, R, MATLAB, Java, MySQL, HTML, JavaScript, Kotlin, Git
Prototyping: Arduino, Raspberry Pi, IoT Sensors, AdaFruit, SolidWorks, AutoCAD, Fusion, EasyEDA, Altium, Logisim
Equipment: Fluorescence microscope, Optical spectrometer, Microscope cameras, Tunable filters, Dichroics, Centrifuge, Gel electrophoresis, Thermocycler, Piezo stages, Oscilloscope, Network analyzer, FPGAs, Reflow oven, 3D printers
Network Interfaces: LoRaWAN, Wi-Fi, Bluetooth LE, NFC/RFID, USB, Ethernet, MQTT, I^2C, SPI, UART, CAN

Research Experience

Opto-Dye: Virtual Cell Staining, *Wheeler Microfluidics Lab (Supervisor: Prof. Aaron Wheeler)* Mar 2025 – Present
- Selected as one of 25 Laidlaw Research Scholars and awarded **$12,500** in funding for translational microfluidics research
- Collaborated with the Ontario Centre for Forensic Sciences to develop a microfluidic–optical system for DNA analysis
- Designed an adaptive LED illumination system enabling low-cost hyperspectral microscopy across 350-1080 nm
- Developed a deep-learning cell classifier using **Detectron2** and μ**SAM** for unstained multi-channel microscopy samples
- Created a portable, low-power imaging unit 50 times cheaper than commercial systems with comparable precision
- **Publication Link:** https://doi.org/10.7490/f1000research.1120482.1

ChromaFlash, *Computational Health & Interaction Lab (Supervisor: Prof Alex Mariakakis)* Apr 2024 – Present
- Built an adaptive illumination system to enhance optical contrast between materials via a programmable LED flash
- Spectrally characterized over 50 materials across 400-900 nm using an optical spectrometer to get absorption features
- Constructed high-speed PCBs using **Altium** to support 7 ns LED switching speed for precise illumination
- Implemented per-channel brightness and timing control on a Raspberry Pi Pico 2 using **C++** and **MicroPython**
- Created a gradient-based optimization algorithm to generate illumination patterns for maximum spectral separability

Cognitive Restructuring, *Intelligent Adaptive Interventions Lab (Supervisor: Prof J.J. Williams)* Aug 2024 – Aug 2025
- Selected as one of four undergraduates in the prestigious Machine Learning research program at IAI labs
- Developing an LLM-based framework for cognitive restructuring using LangChain, Keras and TensorFlow in **Python**
- Optimized model architecture for a **40% increase** in inference efficiency and a **30% decrease** in processing latency
- Redesigned a more user-friendly UI using React and Node.js in **Javascript**, increasing the daily user base by **200 users**

AgroArmor, *United Nations Innovation Labs* Apr 2024 – Nov 2024
- Won first place in the United Nation's Transformative Research Challenge and secured **$5000** in research funding
- Designed a scalable, modular biogas plant addressing energy production in rural agricultural settings
- Engineered custom PCBs and IoT sensor modules using **Altium** to implement continuous low-power slurry agitation
- Prototyped hardware using **Fusion 360** and **AutoCAD**, modeled gas flow and power consumption using **MATLAB**
- Integrated eco-friendly materials, reducing costs by **30%** while ensuring durability and repair-ability

Mersivity: Inclusive Immersive Tech, *Mannlabs (Supervisor: Prof Steve Mann)* Oct 2023 – Jan 2024
- Engineered Virtual Reality(VR) tech for Spinal Muscle Atrophy patients using an EEG-based direct brain interface
- Built 3 Sequential Wave Imaging Devices capable of creating a VR environment from a singular LED strip
- Prototyped with a custom PCB made using **EasyEDA** with addressable LED strips controlled using an ESP32 Feather
- Integrated with a Muse InteraXon EEG headband, using **Python** and **C++** to convert sensor readings into VR actions

Work Experience

Founder, *STEMabled - Accessible education initiative* May 2022 – Aug 2023

- Founded a startup to develop and promote inclusive Arduino STEM kits among specially-abled high school students
- Designed and 3-D printed four **Arduino**-based development kits with braille markings using **TinkerCAD**
- Distributed and tested in collaboration with IIT Delhi, Plaksha University, and the Indian Institute for the Blind
- Conducted hands-on **C++** and **MicroPython** workshops for **150+** visually impaired students and trained **16** faculty
- Developed in-depth audio and braille based training modules for kits to promote independent exploration

Research Intern, *Tendril Solutions Research* May 2023 – Jul 2023

- Developed predictive models to forecast valuations of electric vehicles using Keras and TensorFlow in **Python**
- Performed a comparative study to identify which social media platform provides the most accurate brand sentiment
- Improved data collection methods for perception and brand value by parsing online discourse on Reddit and Twitter
- Increased model accuracy to **94%**, achieved an R^2 score of **0.92**, reduced processing time by **18%**, and token cost by **10%**

Clubs and Extracurriculars

UofTHacks, *Sponsorship Executive* May 2025 – Present

Leading sponsor outreach for UofT's flagship hackathon. Contacted 100+ sponsors, pitched initiatives, and secured funding

USRP (Undergraduate Student Research Program), *Research Mentor* Jan 2025 – Present

Organizing workshops and mentoring 10 undergrads in starting projects, building skills, and navigating opportunities

UofT Blueprint, *Software Developer* Sep 2024 – May 2025

Developing an Android app for Period Purse, a menstrual equity charity. Utilizing **Kotlin**, Coroutines, and Jetpack.

UTMIST, *Project Developer* Aug 2024 – Aug 2025

Developing an AI-based Task Automation pipeline for Lovelytics. Utilizing **Python**, Hugging Face and TensorFlow.